

The Physical AI Scope Test: The Tripartite Architectural Criterion

CRITERION 1: LEARNED MODELS

Physical AI exists exclusively at the simultaneous intersection of learned policies, physical feedback, and delegated authority.

Policy acquired inductively
Stochastic neural networks · V
Non-analytical; unspecifiable

Pure Digital ML

Offline LLMs · Image Classifiers

Digital Autonomous Systems

Stress-Test 3: Flash Crash Algorithm

High-Frequency Algorithmic Trading

Autonomous Cloud VM Autoscalers

Learned + automated authority, but operates on discrete databases with transaction rollback ($p=0$).

CRITERION 2: DELEGATED AUTHORITY

actuator registers (DAC, PWM)
in without synchronous human veto
delay converts into distance

Open-Loop Actuation

Dumb Relays · Fixed Timers

Advisory Decision Support

Stress-Test 1: Advisory Observer

Driver Alert / Drowsiness Cam

Medical Diagnostic Vision HUD

Feedback exists, but output is chime/pixels.

Safety defended by human physiology.

PHYSICAL AI

the Causal Boundary

Autonomous Vehicles · Legged
Manipulators · Cyber

holds unmediated motor authority
kinetic momentum, and energy

Classical Control Theory

Stress-Test 2: Deterministic Governor

1 kHz Quadrotor PID Controller

Watt Steam Governor · Thermostat

Direct actuator authority, but specified analytically with Lyapunov stability proofs.

Passive Mechanics

Spring-Mass Dampers · Insulation

The Borderline Case: Shared Autonomy & The Reflex Gap (Stress-Test 4)

In surgical robotics and steer-by-wire assist, human neuromuscular reflex latency introduces an inescapable delay of **150–250 ms**.

If a learned policy can impart un-overridable physical momentum ($\Delta x = \int_0^{\Delta t} v dt$) before human reflexes engage, the system **crosses into Physical AI**.

Sensors observe own mechanical/thermal consequences (o_{t+1})

Governed by mechanics, momentum ($p = mv$), and thermodynamics ($I^2 R$)